

SMART HOME LIGHTING SYSTEM

Automation System Design Report

1. Introduction

A Smart Home Lighting System is an automation-based system designed to control lights automatically using sensors without human intervention. The system improves energy efficiency, reduces electricity wastage, and enhances user comfort.

In this project, we design a system using Arduino Uno, a PIR motion sensor, and an LDR (Light Dependent Resistor). The system turns ON the light only when motion is detected and the surrounding environment is dark.

This type of automation is widely used in modern smart homes, offices, street lighting systems, and security-based applications.

2. Objective of the System

The main objectives of this system are:

- To automatically control lighting based on human presence.
- To reduce unnecessary power consumption.
- To detect motion using a PIR sensor.
- To measure ambient light using an LDR sensor.
- To design a low-cost and efficient automation system using Arduino Uno.
- To simulate and verify the circuit using Tinkercad.

3. Components Used

The system consists of the following components Fig.1:

3.1 Arduino Uno

Arduino Uno is the central controller of the system. It processes input signals from sensors and controls the output device (LED). It is based on the ATmega328P microcontroller and is widely used in embedded systems and robotics projects.

3.2 PIR Motion Sensor (HC-SR501)

The PIR sensor detects human motion by sensing infrared radiation emitted by the human body.

Key Features:

- Operating voltage: 5V
- Detection range: 3–7 meters
- Digital output signal (HIGH/LOW)
- Adjustable sensitivity

When motion is detected, the sensor sends a HIGH signal to Arduino.

3.3 Photoresistor (LDR)

An LDR is used to measure light intensity in the environment.

Working Principle:

- Resistance decreases when light intensity increases.
- Resistance increases in dark conditions.

The LDR is used to determine whether the room is bright or dark.

3.4 10k Ω Resistor

The 10k Ω resistor is used with the LDR to form a voltage divider circuit. This allows Arduino to read analog voltage values from the LDR.

3.5 LED (Light Output)

The LED represents the room light in the simulation. In real-world applications, this would be replaced by a bulb or lighting system.

3.6 220Ω Resistor

The 220Ω resistor is used to protect the LED from excessive current and prevent damage.

3.7 Jumper Wires

Jumper wires are used to connect all components together in the circuit.

4. Circuit Design (Tinkercad Simulation)

The circuit is designed using Tinkercad with the following connections Fig.2:

PIR Sensor:

- VCC → 5V (Arduino Uno)
- GND → GND (Arduino Uno)
- OUT → Digital Pin 2 (D2)

LDR Voltage Divider:

- One leg of LDR → 5V
- Other leg of LDR → A0
- 10kΩ resistor between A0 and GND

LED Connections:

- Digital Pin 8 → 220Ω resistor
- Resistor → LED anode (+)
- LED cathode → GND

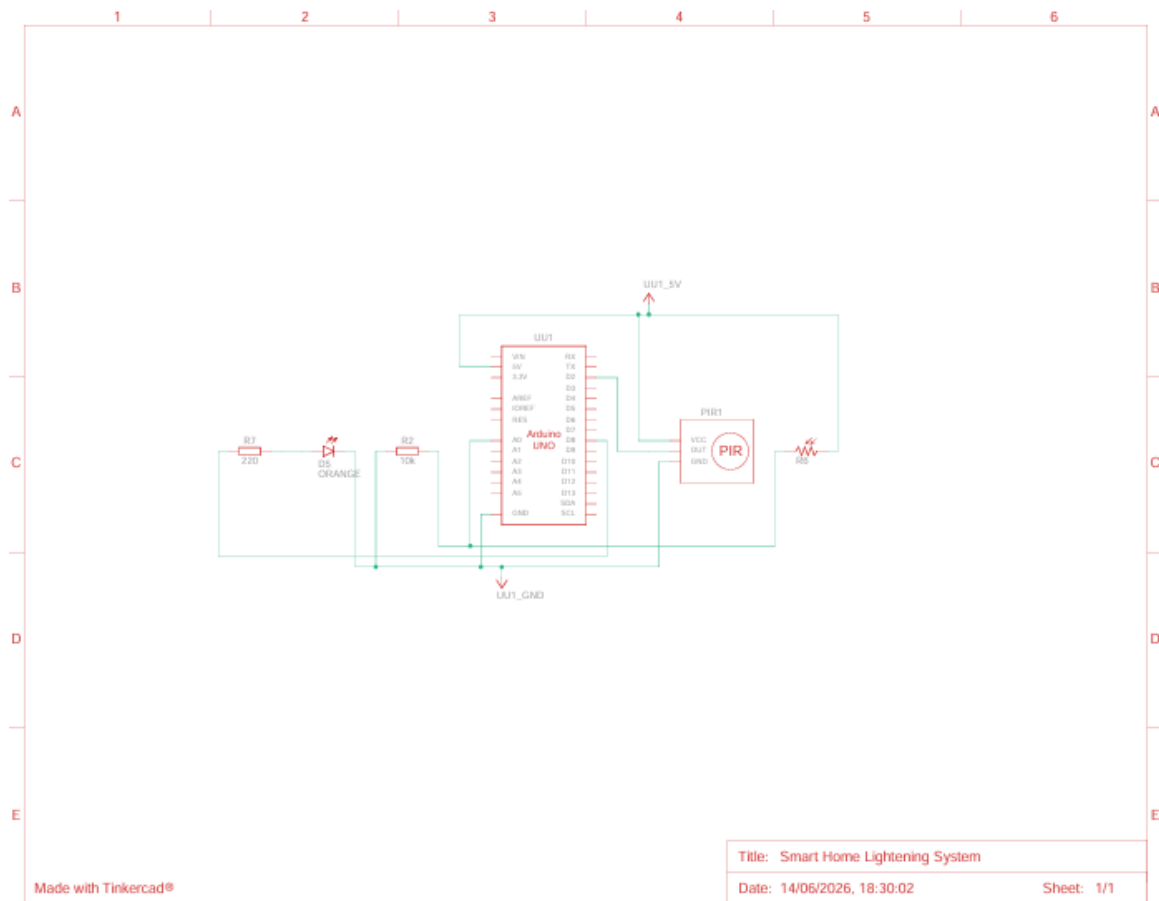


Fig.1: shows Schematic Design

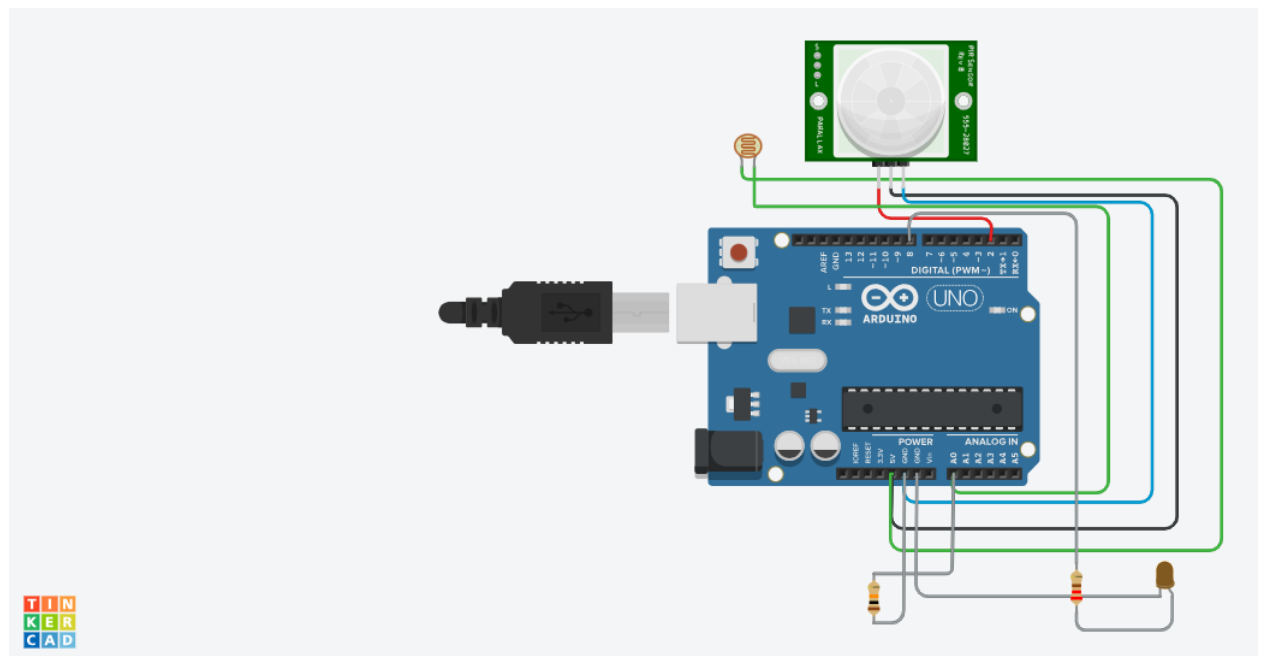


Fig.2: shows Circuit Design

5. Working Principle

The system operates in the following steps:

1. The PIR sensor continuously monitors motion.
2. The LDR sensor measures ambient light intensity.
3. Arduino reads both sensor values.
4. Decision logic is applied:
 - If motion detected AND dark → LED ON
 - Otherwise → LED OFF
5. The process repeats continuously.

6. Circuit Logic

Motion Detected = HIGH (PIR)

Dark Environment = Low LDR Value

Condition for ON:

PIR = HIGH AND LDR < Threshold

Otherwise, LED remains OFF

11. Conclusion

The Smart Home Lighting System demonstrates an efficient automation solution using Arduino Uno, PIR sensor, and LDR sensor. The system successfully automates lighting based on motion detection and ambient light conditions. This project helps reduce energy consumption and represents a basic model of modern smart home technology.